

Visual Product Search

Pipeline (per project spec):

1. **YOLO** — detect and crop the clothing item from the query image
2. **BLIP-2 caption** — generate a caption for the query crop
3. **CLIP (fused)** — encode crop + caption, fuse with alpha weighting
4. **HNSW ANN** — retrieve top-K×{rerank} candidates by cosine similarity
5. **BLIP ITM re-ranking** — score each candidate's pre-computed caption against the query image, re-rank to final top-K

Upload a query image

 14_1_front.jpg
13.5KB

+

Original Image

YOLO Detection



2 clothing box(es) found — 0.04s

Select / Confirm Crop for Search

Individual clothing crops:

Search Settings

Top-K results

10

Re-rank pool multiplier (fetch top-K × N before re-ranking) ?

5

Fusion alpha (α -image + (1- α)-caption) ?

0.50

☒ Show similarity & ITM scores

☒ Show candidate captions

Index Info

Index: `hnsw_alpha0.5_seed51.bin`

Fusion α : `0.5`

Re-rank pool: top-K × 5 = top-50

Pipeline steps:

1. Upload image
2. YOLO detects clothing boxes
3. User selects / confirms crop
4. BLIP-2 generates caption for crop
5. CLIP encodes crop + caption, fused at α =0.5
6. HNSW retrieves candidate pool
7. BLIP-2 ITM re-ranks by image↔caption score

Device: `cuda`



Crop 1 · clothing (conf=0.96, 81×170)



Crop 2 · clothing (conf=0.76, 85×34)

Full-body crop (union of all clothing regions):



Full-body crop · 95×201

Confirm crop to search with:

- ☒ Crop 1 · clothing (conf=0.96)
- ☐ Crop 2 · clothing (conf=0.76)
- ☐ Full-body crop (95×201)

Final Crop



81×170

 Search with this crop

Running Query Pipeline

Step 2 done — Caption: "men's tank top" (5.55s)

Step 3 done — Fused embedding $\alpha=0.5$ (0.05s)

Step 4 done — Retrieved 50 candidates (0.00s)

Step 5 — Re-ranking 50 candidates with BLIP-2 ITM (query image ↔ candidate caption)...

Re-ranking done (0.00s)

Step 5 done — Re-ranked to top-10 (0.00s)

Pipeline Timings

Step 1 — YOLO	Step 2 — BLIP-2 C...	Step 3 — CLIP Fused	Step 4 — HNSW	Step 5 — ITM Re-r...
0.04s	5.55s	0.05s	0.00s	0.00s
Total Pipeline Time				
5.64s				

Top-10 Results (after ITM Re-ranking)

Fused query $\alpha=0.5$ · Pool: top-50 by cosine sim → re-ranked by BLIP-2 ITM

				
Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
Item: id_00005955	Item: id_00004983	Item: id_00005955	Item: id_00005955	Item: id_00004530
ITM: 0.000	ITM: 0.000	ITM: 0.000	ITM: 0.000	ITM: 0.000
Cosine: 0.884	Cosine: 0.881	Cosine: 0.873	Cosine: 0.869	Cosine: 0.858
				
Rank 6	Rank 7	Rank 8	Rank 9	Rank 10
Item: id_00004983	Item: id_00005160	Item: id_00004698	Item: id_00004698	Item: id_00004028
ITM: 0.000	ITM: 0.000	ITM: 0.000	ITM: 0.000	ITM: 0.000
Cosine: 0.856	Cosine: 0.855	Cosine: 0.854	Cosine: 0.851	Cosine: 0.851

